

Concern-Weighted Weakness

Subtitle. A bridge theorem from Bennett extension to structure-compatible generalization and load-bearing concern.

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Abstract

Bennett's weakest-hypothesis principle defines the weakness of a statement as the cardinality of its extension: weaker hypotheses rule out fewer possible decisions while still fitting the observed child task. Recent structure-compatible generalization experiments operationalized weakness differently, as the number of deployment-relevant transformations under which a learned function remains compatible. This note proves the bridge between the two quantities. Group-compatible weakness is Bennett weakness restricted to a transformation-generated deployment slice, under explicit block-decomposition assumptions. We then introduce concern-weighted weakness: extension mass weighted by consequence, viability, value, cost, or future commitment importance. Unweighted Bennett weakness is the special case in which every future decision has equal concern; group weakness is the special case in which the deployment slice decomposes into equally weighted transformation blocks; concern-weighted group weakness is the natural selector when OOD futures have unequal consequence. Finally, under a standard finite-capacity rate-distortion approximation, representation density is predicted to allocate according to consequence-weighted compatible extension mass. The paper separates what is proved from what is empirical: existing OOD, semantic-selection, metric-deformation, vector-concern, re-engagement, and topology-negative results are evidence anchors and boundary conditions, not assumptions smuggled into the theorem.

1. Motivation

The current research program has three pieces that were previously adjacent but not formally connected.

First, Bennett's weakest-hypothesis theorem argues against a purely shortest-description account of generalization. In Bennett's formalism, a hypothesis is weak when its extension is large. It says little while still satisfying the observed task, and therefore remains compatible with more possible parent tasks under a uniform task prior.

Second, the structure-compatible generalization results measure a finite, checkable version of weakness for functions: count the transformations under which a train-consistent candidate remains equivariant up to an output action. In cyclic, dihedral, rotated-stroke, and semantic retrieval settings, this compatibility score predicts or selects better OOD behavior than train loss or in-distribution validation when the deployment shift is generated by the measured structure.

Third, the concern-geometry and load-bearing concern results say that not all extensions matter equally. Finite systems cannot preserve every difference. They allocate resolution, inquiry, memory, and action-binding toward distinctions with consequence. Concern is therefore not introduced here as phenomenology or feeling. It is a nonnegative weighting over future decisions, states, or commitments.

The bridge is simple:

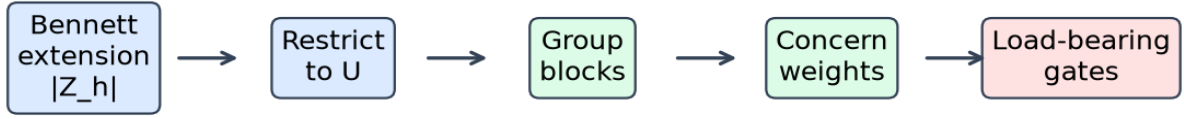
Bennett weakness = unweighted extension mass.

Group weakness = extension mass on a group-generated OOD slice.

Concern-weighted weakness = extension mass weighted by future consequence.

Load-bearing concern = weighted extension that remains identifiable and reaches commitment.

The point of this paper is to make that bridge theorem precise enough to be useful and narrow enough to be honest.



The first four steps are proved; load-bearing gates are the formal evaluation filter.

Theorem ladder

2. Setup

Let L be a finite implementable language or finite decision universe. For a hypothesis or statement h , let $Z_h \subseteq L$ denote its extension: the set of decisions, completions, or statements still permitted by h .

Let α be the observed child task and ω an unknown parent task. A candidate hypothesis is admissible if it is a model of the observed task:

$$h \in M_\alpha.$$

Let $U \subseteq L$ be the deployment slice: the decisions not settled by the observed child task but relevant to the parent task, OOD split, memory delay, tool interface, or future commitment surface.

The deployment-restricted extension of h is:

$$Z_h^U = Z_h \cap U.$$

Define restricted Bennett weakness:

$$\text{Weak}_U(h) = |Z_h^U|.$$

This is not the full Bennett weakness $|Z_h|$. It is the portion of the extension that falls inside the deployment region whose coverage we care about.

Finally, define concern as a nonnegative measure over deployment decisions:

$$\kappa: U \rightarrow \mathbb{R}_{\geq 0}.$$

$\kappa(z)$ can represent consequence, viability value, cost of error, priority weight, future action relevance, or commitment importance. The mathematics only requires nonnegativity. The interpretation depends on the task.

Concern-weighted weakness is:

$$\text{Weak}_{\{U, \kappa\}}(h) = \sum_{z \in Z_h^U} \kappa(z).$$

When $\kappa(z)=1$ for all z , this reduces to deployment-restricted Bennett weakness.

3. Theorem 1: Deployment-Restricted Bennett Weakness

Theorem 1. Among train-consistent hypotheses, maximizing deployment-restricted Bennett weakness maximizes covered deployment extension under a uniform decision-level utility over U .

Assumptions.

- The candidate set H_α contains only hypotheses that fit the observed child task.
- A deployment decision $z \in U$ is counted as covered by h exactly when $z \in Z_h$.
- Each deployment decision has equal utility.

Claim.

$$h^* \in \arg\max_{\{h \in H_\alpha\}} \text{Weak}_U(h)$$

maximizes the number of covered deployment decisions.

Proof.

By definition, $\text{Weak}_U(h) = |Z_h \cap U|$. Under assumption 2, $Z_h \cap U$ is exactly the set of deployment decisions covered by h . Under assumption 3, total utility is the cardinality of that set. Therefore any maximizer of Weak_U maximizes covered deployment utility. QED.

This theorem is intentionally modest. Bennett's original proof concerns probability of generalization under a uniform distribution over tasks in his formalism. Here we isolate the deployment slice because the structure-compatible experiments measure OOD coverage, not the entire implementable language.

4. Theorem 2: Concern-Weighted Weakness

Theorem 2. If deployment utility is additive and nonnegative over future decisions, then maximizing concern-weighted weakness maximizes expected concern-weighted deployment coverage among train-consistent hypotheses.

Assumptions.

- H_α contains train-consistent hypotheses.
- A deployment decision $z \in U$ is covered by h exactly when $z \in Z_h$.
- Deployment utility is additive over covered decisions:

$$\text{Utility}(h) = \sum_{z \in Z_h \cap U} \kappa(z),$$

with $\kappa(z) \geq 0$.

Claim.

$$h^* \in \arg\max_{h \in H_\alpha} \text{Weak}_{\{U, \kappa\}}(h)$$

maximizes deployment utility.

Proof.

The definition of concern-weighted weakness is identical to the assumed deployment utility:

$$\text{Weak}_{\{U, \kappa\}}(h) = \sum_{z \in Z_h \cap U} \kappa(z) = \text{Utility}(h).$$

Thus any maximizer of $\text{Weak}_{\{U, \kappa\}}$ maximizes deployment utility. QED.

This theorem is almost tautological, but useful: it shows exactly where "concern" enters. Concern is not a new metaphysical primitive inside the proof. It is the weighting measure that decides which parts of a hypothesis extension matter for future performance.

5. Theorem 3: Group-Compatibility Bridge

The structure-compatible generalization papers use a group-style quantity:

$$C_G(h) = \{g \in G : h \text{ is compatible with } g\}.$$

For a learned function f , compatibility typically means there exists an output-side action such that:

$$f(g \cdot x) = h_g \cdot f(x)$$

for the relevant domain points.

We now connect this to Bennett extension.

Theorem 3. Suppose the deployment slice decomposes into transformation blocks. If a hypothesis covers a block exactly when it is compatible with the corresponding transformation, then concern-weighted Bennett weakness on the deployment slice equals a weighted count of compatible transformations.

Assumptions.

- G is a finite transformation set.
- The deployment slice decomposes as a disjoint union:

$$U = \text{disjoint_union}_{g \in G} B_g.$$

- For each g , there is a block $B_g \subset U$ of deployment decisions generated by applying g to observed or otherwise anchored cases.

- Block coverage is equivalent to compatibility:

$$B_g \subset Z_h \text{ iff } g \in C_G(h).$$

- Partial block coverage is absent or ignored at this abstraction level:

$$\text{if } g \notin C_G(h), \text{ then } (Z_h \cap B_g) = \text{empty}.$$

Define block concern mass:

$$\mu_g = \sum_{z \in B_g} \kappa(z).$$

Claim.

$$\text{Weak}_{\{U, \kappa\}}(h) = \sum_{g \in C_G(h)} \mu_g.$$

Proof.

Start from the definition:

$$\text{Weak}_{\{U, \kappa\}}(h) = \sum_{z \in Z_h \cap U} \kappa(z).$$

Using the disjoint block decomposition:

$$Z_h \cap U = \text{disjoint_union}_{\{g \in G\}} (Z_h \cap B_g).$$

Therefore:

$$\begin{aligned} \text{Weak}_{\{U, \kappa\}}(h) \\ = \sum_{g \in G} \sum_{z \in Z_h \cap B_g} \kappa(z). \end{aligned}$$

By assumptions 4 and 5, the inner sum is μ_g when $g \in C_G(h)$ and 0 otherwise. Thus:

$$\text{Weak}_{\{U, \kappa\}}(h) = \sum_{g \in C_G(h)} \mu_g.$$

QED.

Corollary 3.1: unweighted group weakness. If $\kappa(z)=1$ and all blocks have equal size $|B_g|=c$, then:

$$\text{Weak}_U(h) = c * |C_G(h)|.$$

Therefore:

$$\text{argmax}_h \text{Weak}_U(h) = \text{argmax}_h |C_G(h)|.$$

This is the exact bridge: group-compatible weakness is Bennett weakness restricted to an equally weighted, transformation-generated deployment slice.

Corollary 3.2: concern-weighted group weakness. If block masses are unequal, the correct selector is:

$$h^* \in \text{argmax}_h \sum_{g \in C_G(h)} \mu_g.$$

Unweighted group weakness can prefer a hypothesis that covers many low-consequence transformations over one that covers fewer high-consequence transformations. Concern weighting repairs that when the deployment utility is known or estimable.

6. Corollary 4: Finite-Capacity Metric Allocation

Concern-weighted weakness is a selector over hypotheses. The metric-deformation papers ask a representation question: where should a finite code allocate resolution?

Let x index a task state, semantic region, spatial location, or commitment-relevant condition. Let:

$$\begin{aligned} \tau_h(x) &= \text{compatible extension mass reaching } x \\ \kappa(x) &= \text{consequence weight at } x \\ \lambda_h(x) &= \kappa(x) * \tau_h(x). \end{aligned}$$

$\lambda_h(x)$ is consequence-weighted compatible extension pressure. In words: how much future-relevant compatible continuation mass concentrates at x .

Assume a high-resolution finite-capacity code with local representational density $\rho(x)$, fixed capacity:

$$\int \rho(x) dx = C,$$

and local distortion:

$$D(x) = K * \rho(x)^{(-2/d_{\text{eff}})}.$$

The weighted distortion objective is:

$$L[\rho] = \int \lambda_h(x) * K * \rho(x)^{(-2/d_{\text{eff}})} dx$$

subject to the capacity constraint.

Corollary 4. Under these assumptions, the minimizing allocation obeys:

$$\rho(x) \text{ proportional to } \lambda_h(x)^{d_{\text{eff}}/(d_{\text{eff}}+2)}.$$

Equivalently:

$$\rho(x) \text{ proportional to } [\kappa(x) * \tau_h(x)]^{d_{\text{eff}}/(d_{\text{eff}}+2)}.$$

Proof sketch.

Minimize the Lagrangian:

$$\int K * \lambda_h(x) * \rho(x)^{(-2/d_{\text{eff}})} dx + \eta * \int \rho(x) dx.$$

The first-order condition is:

$$(-2/d_{\text{eff}}) * K * \lambda_h(x) * \rho(x)^{-(d_{\text{eff}}+2)/d_{\text{eff}}} + \eta = 0.$$

Solving for $\rho(x)$ yields:

$$\rho(x) \text{ proportional to } \lambda_h(x)^{d_{\text{eff}}/(d_{\text{eff}}+2)}.$$

QED.

This is an idealized allocation corollary, not a claim that every trained network will realize the nominal physical dimension. The existing effective-dimension experiment is exactly the right caution: in the tested path-integration harness, the hoped-for two-dimensional exponent was rejected and the measured exponent was closer to an effective one-dimensional bottleneck.

7. Load-Bearing Concern as a Filter

The theorems above connect extension, transformation, and consequence. The load-bearing concern doctrine adds an evaluation rule: a distinction should count as agentially possessed only if it survives the route to behavior.

This can be represented as a filter on κ .

Let:

$s(z)$ in $\{0,1\}$ = selected under concern
 $p(z)$ in $\{0,1\}$ = preserved across the relevant delay or transformation
 $i(z)$ in $\{0,1\}$ = identifiable against proxy structures
 $r(z)$ in $\{0,1\}$ = reopenable when contact with the world becomes stale
 $c(z)$ in $\{0,1\}$ = committed at the relevant action, tool, speech, repair, query, or refusal surface

Define load-bearing concern:

$$\kappa_{\text{LB}}(z) = \kappa(z) * s(z) * p(z) * i(z) * r(z) * c(z).$$

Then:

$$\text{Weak}_{\{U, \kappa_{\text{LB}}\}}(h) = \sum_{z \text{ in } Z_h \text{ cap } U} \kappa_{\text{LB}}(z).$$

This does not prove consciousness, subjective concern, or full agency. It formalizes the evaluation discipline used throughout the program: behavior alone is not enough; the weighted extension only counts when the intended structure is selected, preserved, proxy-resistant, reopenable, and committed.

The filter also explains why failed gates matter. If a model carries a decodable variable that does not affect the tool call, then $c(z)=0$. If a shortcut passes the benchmark without the intended invariant, then $i(z)=0$. If the system preserves a stale memory and cannot re-engage after world change, then $r(z)=0$. The extension may be large, but the load-bearing extension is small.

8. Empirical Anchors

The paper's proofs are independent of the experiments, but the experiments motivate the definitions and bound the claim.

Structure-compatible OOD. In finite cyclic and dihedral tasks, the symmetry-compatible selector recovers the invariant rule in 100% of trials while tested loss, validation, description-length, MDL-style, compression, and flatness proxies recover it in 0% of those settings. In neural cyclic sweeps, learned-function weakness under the true group remains the strongest OOD correlate at 4096 models, with Pearson $r=+0.8085$ and a fixed-effect residual signal after controlling augmentation condition.

Semantic model selection. The semantic-selection phase turns compatibility into an OOD-free selector among finite retrieval candidates. Learned compatibility selects mean OOD 0.978, compared with 0.919 for ID/train/random selectors and 0.751 for wrong compatibility. This is evidence that the group bridge generalizes into learned finite semantic transformations, not only hand-built cyclic groups.

Concern deforms metric. The spatial metric-deformation sweep injects a movable loss-weight field and tests whether learned metric density follows it. Across RNN, Transformer, and JEPA-style spatial models, moved-location lift and specificity are positive. The correct interpretation is narrow: externally specified priority reallocates local representational resolution in controlled finite spatial learners.

Vector concern. The valence-tapestry experiment shows why scalar concern is not enough. Vector Delta-V heads support zero-shot reweighting when internal priority weights shift; scalar drive heads collapse distinctions that later matter. This is an empirical motivation for κ as a multidimensional or context-dependent weight, rather than a single static reward.

Re-engagement and commitment surfaces. The Suite C and long-horizon commitment-surface suites support the load-bearing filter. They ask whether future-critical or stale-world variables reach the action, probe, schema, tool, repair, or generated-value surface, not merely whether they are present somewhere in hidden state.

Topology boundary. The grid-cell topology result is an important negative. Translation augmentation produces toroidal, larger-arena-generalizing codes, and weakness tracks one spectral aspect of learned translation structure. But weakness does not govern torus formation or mediate the OOD relation in that harness. This is exactly what the bridge theorem predicts it should be possible to find: compatibility mass is useful when the measured compatibility family is the load-bearing deployment structure, but it can be a footprint rather than the cause.

9. What Is Proved and What Is Not

Proved.

- Deployment-restricted weakness is extension mass on the deployment slice.
- Concern-weighted weakness is the additive weighted extension objective.
- Group-compatible weakness is a special case of restricted Bennett weakness when deployment futures decompose into equally weighted transformation blocks.
- Concern-weighted group weakness is the correct weighted selector under unequal block consequence.
- The finite-capacity allocation corollary follows from the stated rate-distortion assumptions.

Not proved.

- That all intelligence is governed by concern-weighted weakness.
- That a neural network will discover the correct G , U , or κ without inductive bias or supervision.
- That representation density in real models always follows the nominal exponent.
- That load-bearing concern implies consciousness or subjective experience.
- That compatibility is causal rather than diagnostic in every system.

The safe claim is narrower:

When deployment uncertainty is structured by transformations,
and future decisions have unequal consequence,
the relevant weakest valid hypothesis maximizes
consequence-weighted compatible extension.

10. A Targeted Experiment to Add Next

A direct next experiment should make the concern weighting load-bearing.

Construct a train/ID-tied structured OOD benchmark where missing cases have unequal consequence weights. Compare:

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train selector
ID selector
unweighted group weakness
concern-weighted group weakness
wrong-weight control
wrong-group control
oracle OOD selector
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Prediction:

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concern-weighted weakness wins on weighted OOD utility;
unweighted weakness may win on raw OOD accuracy;
wrong-weight and wrong-group controls fail.
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This experiment would test the part that is new here. Existing OOD work supports the group bridge; existing metric-deformation work supports concern-based allocation; the new result should force a selector to choose between many low-value compatible futures and fewer high-value compatible futures.

11. Discussion

The bridge is useful because it clarifies the relationship between three vocabularies that otherwise sound unrelated.

Bennett's theorem says that the best valid hypothesis is the weakest, not necessarily the shortest. The structure-compatible work says that, in group-generated OOD settings, weakness can be measured as compatible transformation volume. The concern doctrine says that finite systems should not preserve all extensions equally, because not all future distinctions carry equal consequence.

Put together:

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Finite systems should preserve the weakest valid structure
over future transformations that matter,
weighted by consequence,
and counted only where the structure remains load-bearing.
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That sentence is not the theorem. It is the research program distilled from the theorem plus the empirical stack. The theorem tells us how to compute the quantity once U , G , and κ are specified. The experiments tell us where the quantity has worked, where it has failed, and what must be tested next.

The practical implication is straightforward. When train and ID validation tie, ask:

- What future deployment slice matters?
- Which transformations generate that slice?
- Which parts of the slice have high consequence?
- Which candidate hypotheses preserve the most consequence-weighted compatible extension?
- Does the preserved structure actually reach the commitment surface, or did a proxy pass the metric?

This turns concern from a poetic label into a weighting measure over Bennett extension, and turns weakness from a philosophical prior into a finite model-selection and evaluation object.

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